
ECE175B Project Proposal

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1 Proposal

1. What research problem are you aiming to solve?

In the recent years, the rise of AI has led to highly sophisticated and intelligent Large Language models - LLMs capable of forming natural and human like language. LLMs have also shown an improvement over traditional machine translations as they are able to incorporate and understand contextual and relational information much better than traditional methods. Gao et al. [2024] However, LLMs have yet to master the translation of foreign language poems, which is far more difficult than academic or everyday language. As Robert Frost puts it, "Poetry is what gets lost in translation," which implies that poetry is some intangible, ineffable thing, which although constructed in language cannot be transposed across languages Gao [2010-01-01]. We aim to solve the problem of efficient and high-quality English translations of classical Chinese poetry with small or medium sized models. The challenge is not only to preserve the poem's literal meaning, but also to retain its cultural references, imagery, conciseness, and poetic style in English. Our project studies whether parameter-efficient fine tuning can make compact models competitive on this task.

2. Why is this problem important?

Classical Chinese poetry is linguistically difficult to translate due to its highly compressed, heavily allusive, and poetic features. A high-quality translation must not only be semantically correct but should also sound natural and retain its original poetic quality which only highly skilled translators are able to replicate in the English language. On top of that, current existing LLMs struggle to mimic poetic language to the extent that it can for academic and everyday language. This limits the resources students and researchers who rely on accurate English translation to study Chinese literature.

3. Discuss existing works that aim to solve this problem and their limitations.

The main problem is that while general-purpose LLMs produce fluent literal translations, they often fail to preserve adequacy, meaning and form. Recent research on LLMs, such as ChatGPT, shows improved accuracy over traditional translation architectures like Google Translate and DeepL Gao et al. [2024]. However, because these large models are trained on overwhelmingly generic internet data, they still fall short of preserving appropriate historical context, concise expression, and poetic elegance. This can be attributed to the lack of existing datasets of high quality Chinese to English translations of poetry. Chen et al. [2025]

To address these limitations, Chen et al. proposed RATs, a fully retrieval-augmented pipeline as well as PoetMT, a benchmark that includes contextual information from external databases to provide richer background for LLM translation Chen et al. [2025]. While RATs effectively improved translation, the pipeline depended on heavy inference making it complex and

costly to deploy. Another relevant line of work is PoetryQwen, which fine-tunes Qwen2.5-14B using LoRA and builds a domain-specific dataset, CCPoetry-49K. Its results show that LoRA can significantly improve performance on classical poetry understanding tasks such as term interpretation, semantic interpretation, and emotional inference. However, its task is mainly poetry appreciation and understanding, not direct Chinese-to-English poetry translation Xie [2025].

4. Your proposed solution for solving this problem

Addressing the existing limitations, the more meaningful direction is to move RAT’s knowledge supplementation ability into the training stage, while keeping LoRA’s low-cost adaptation ability at deployment time.

We propose a two path framework for classical Chinese poetry translation. Our baseline is a lightweighted decoder-only model. We use Qwen2.5-1.5B and apply QLoRA to adapt it efficiently. The training data will combine: a small amount of high-quality Chinese-English classical poetry translation data and NiuTrans / CCLUE / CCPM-style auxiliary understanding data for semantic support Xie [2025].

Our second path is mT5-base + encoder-decoder LoRA. Since mT5-base is architecturally more translation-oriented, this path lets us test whether an encoder-decoder model is inherently better suited for poetry translation than a decoder-only model of similar scale Xue et al. [2021]. This comparison is important because it separates the effect of architecture choice from the effect of parameter-efficient adaptation.

5. Novelty and significance of your solution

The primary novelty of our solution lies in framing translation not just as a task of linguistic equivalence, but of stylistic cloning via parameter-efficient fine-tuning. By studying both Qwen2.5-1.5B + QLoRA and mT5-base + LoRA we can compare which route works better. This project is significant because it provides a scalable, low-compute framework for Chinese poetry translation. If successful, this methodology could be easily adapted to other low-resource or highly stylized translation tasks, such as translating ancient prose, religious texts, or indigenous storytelling.

6. Tentative timeline

- **Week 4:** Literature review, dataset collection (cleaning parallel poetry datasets), and baseline evaluation of the pre-trained SLM.
- **Weeks 5-6:** Compile Midterm Report, environment setup, path 1: Qwen2.5-1.5B + QLoRA implementation, and initial model fine-tuning runs; add auxiliary data and evaluate improvements(optional).
- **Weeks 7-8:** Implement path 2: mT5-base + LoRA; add auxiliary data and compare both models(optional).
- **Weeks 9:** Quantitative evaluation (BLEU, ROUGE, BERTScore) and qualitative evaluation (human review for elegance).
- **Weeks 10:** Finalize comparisons, draw conclusions, and write the Final Report.

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